

Mixed Membership Estimation in Two-Path Partial Correlation Network Model

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Abstract

We develop a two-path mixed membership partial correlation network model (2PMMPN) for panels of time-series data. The model allows the cross-sectional dependence structure to be driven jointly by mixed membership and node-specific sociability. We first design a two-path mixed membership model using Bayesian network and then impose latent network structure on concentration structure of the panel. The resulting partial-correlation network separates sparsity structure from connection intensity through two distinct latent pathways. We propose an algorithm for estimating the mixed-membership structure, prove consistency of the estimator, and document its finite-sample performance in simulation studies. An empirical application to U.S. state-level employment growth rate illustrates how the method can uncover overlapping latent groups in economic panel data.

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1 Introduction

Latent group structure in panel data has received increasing attention in recent econometric research. One strand of the literature estimates grouped heterogeneity in panel models, allowing parameters to differ across groups while imposing homogeneity within each group, for example in fixed effects or slope coefficients (Bonhomme & Manresa 2015, Su et al. 2016). A second strand recovers group structure from the interdependence patterns of large multivariate systems by imposing community or block structure (Guðmundsson & Brownlees 2021, Chen et al. 2023, Zhu et al. 2023). These approaches are powerful, but they typically rely on pure membership: each unit is assigned to exactly one latent group. This restriction may be inappropriate in many economic and social systems, where units often belong to multiple latent groups simultaneously. For example, firms may be linked to several ownerships, supply-chain, or industry networks; voters may shift across political parties; individuals may have multiethnic backgrounds; and researchers often span multiple disciplines. We address this limitation by developing a method for estimating mixed-membership structure in large panels of time series through the partial-correlation network induced by the concentration matrix of the underlying multivariate system.

In many real-world networks, vertices exhibit group structure: link formation depends systematically on whether two vertices belong to the same latent group. This empirical regularity is commonly referred to as community structure or clustering. A

widely used model for such networks is the degree-corrected stochastic block model (DC-SBM) ([Karrer & Newman 2011](#)). The DC-SBM is a random graph model in which vertices are assigned to latent groups, and the probability of an edge depends on group membership as well as node-specific degree heterogeneity. In the canonical assortative case, vertices in the same group are more likely to be connected than vertices in different groups.

Recent work has introduced community structure into large panels of time series by assuming that the series are partitioned into latent groups and that dependence is stronger within groups than across groups ([Brownlees et al. 2022](#), [Guðmundsson & Brownlees 2021](#), [Guðmundsson 2026](#)). These approaches impose latent group structure either on the partial-correlation network or on a large autoregressive system, typically through a degree-corrected stochastic block model. Although this provides a useful framework for modeling grouped dependence in high-dimensional panels, it has two limitations for our purpose. First, each series is assigned exclusively to one group, ruling out overlapping or mixed-membership patterns. Second, the DC-SBM primarily captures the sparsity pattern of the dependence network, rather than separately modeling the intensity of nonzero conditional dependence.

Motivated by recent developments in mixed-membership network models ([Airolidi et al. 2008](#), [Fan et al. 2022](#)), we first propose a novel two-path mixed membership network model (2PMMM). The model extends traditional MMM by decomposing generative process into two latent pathways. The first pathway determines the

sparsity structure which corresponds to traditional MMM in which a random graph defined over a fixed number of vertices builds links through independent Bernoulli trial. A latent mixed membership structure controls that the vertices are more likely to connect in the same groups than between different groups. Compared with pure membership structure that each vertex belongs to the one group exclusively, mixed membership allows each vertex to affiliate with several groups. The second independent path determines the connection intensity based on the sparsity structure in first path. A latent sociability structure controls the connection intensity under the assumption that the interaction between two more sociable vertices is more likely to lead to high connection weight if the link exists determined in first path. Two paths are combined into a weighted directed network using a joint distribution specified by Bayesian belief network.

We then introduce a two-path mixed membership partial correlation network (2PMM-PCN) in which the dependence structure of a set of series is determined by our proposed 2PMMM. More concretely, the concentration of the multivariate system is a function of the realization of our 2PMMM. The key aspect of the 2PMMPCN is designed so that the panel inherits the latent mixed membership and sociability structures through innovation process of the concentration.

To estimate the latent mixed-membership structure, we develop PartialCorr-Mixed-SCORE, a spectral algorithm that extends SCORE ([Jin 2015](#)) and Mixed-SCORE ([Jin et al. 2024](#)) to high-dimensional panel time-series settings. Unlike the origi-

nal SCORE framework, which recovers pure- or mixed-membership communities from binary adjacency matrices generated by degree-corrected stochastic block models or mixed-membership network models, our method operates on the sample concentration matrix estimated from large panels. The algorithm first applies eigen-ratio normalization to remove node-specific heterogeneity and sociability effects. It then exploits the simplex geometry of Mixed-SCORE to recover mixed-membership weights from the normalized spectral representation.

We establish consistency by adapting the theoretical framework of [Rohe et al. \(2011\)](#). The analysis proceeds in two steps. First, we show that the algorithm recovers the mixed-membership structure at the population level under the model assumptions. Second, we derive bounds on the deviation between the sample and population mixed membership, thereby quantifying the effect of sampling noise on membership estimation. We further conduct simulation studies to assess the finite-sample performance of the PartialCorr-Mixed-SCORE algorithm.

We illustrate the empirical relevance of the proposed method using U.S. state-level employment growth data. The same dataset was analyzed by [Hamilton & Owyang \(2012\)](#) to infer posterior distributions of group memberships and by [Brownlees et al. \(2022\)](#) to detect pure-membership groups. We instead apply PartialCorr-Mixed-SCORE to estimate mixed memberships, allowing each state to partially affiliate with multiple latent groups. The resulting membership patterns share some features with earlier findings but also reveal important differences, suggesting that mixed

membership provides a complementary view of the latent dependence structure underlying U.S. employment dynamics.

This paper contributes to four strands of the literature. First, it contributes to work on latent group structure in panel data ([Bonhomme & Manresa 2015](#), [Su et al. 2016](#), [Brownlees et al. 2022](#), [Guðmundsson & Brownlees 2021](#), [Zhang et al. 2019](#)). Relative to this literature, we estimate mixed-membership structure rather than pure-membership group assignments. Second, the paper relates to mixed-membership network models and their estimation ([Airoldi et al. 2008](#), [Jin et al. 2024](#), [Zhang et al. 2020](#)). Our contribution is to develop a two-path framework that models both the sparsity pattern of the graph and the intensity of nonzero connections, whereas standard mixed-membership network models mainly focus on link formation. Third, the paper builds on probabilistic graphical models and Bayesian networks ([Blei et al. 2003](#), [Koller & Friedman 2009](#), [Munro & Ng 2022](#)). We use this framework to specify a joint distribution for the network and covariance matrix. Finally, the paper is related to studies of social effects and dependence structures in panel data ([Chen et al. 2022](#), [Zhu et al. 2019](#)). We contribute by embedding latent social effect into the covariance matrix of the innovation process, thereby providing a new way to model social effects in panel time series.

The rest of the paper is organized as follows. Section 2 presents the methodology, including the main model, estimation algorithm, theoretical results and discussion. Section 3 reports the results of a simulation study. Section 4 is devoted to the

empirical application. Finally, Section 5 concludes. Technical proofs, identification strategy and additional materials are provided in the supplementary materials.

2 Methodology

2.1 The Two-path Mixed Membership Partial Correlation Network

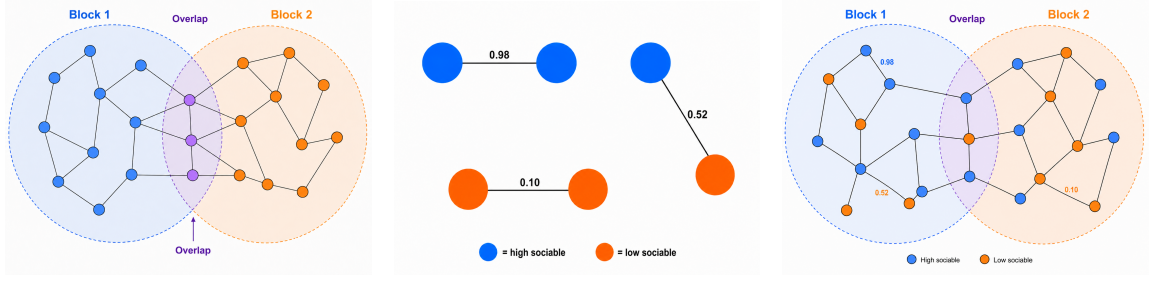
We introduce a model for an (N) -dimensional panel time series $\{Y_t\}_{t=1}^T$, where we have the innovation term ϵ_t as usual. The key feature is that the interdependence structure of the components of $\{Y_t\}_{t=1}^T$ is determined by a latent graph $\mathcal{G}$ endowed with latent structure. The notion is first introduced in [Brownlees et al. \(2022\)](#). The difference between the models in [Brownlees et al. \(2022\)](#) and ours is that we impose a latent mixed membership structure but [Brownlees et al. \(2022\)](#) impose the pure membership structure. In addition, we impose one more sociability structure.

It's well known that the linear partial dependence structure of the system is embedded in the inverse of the covariance Σ . We call the $\Lambda = \Sigma^{-1}$ the concentration, or precision, matrix ([Dempster 1972](#)). Whereas the covariance matrix captures marginal dependence, the concentration matrix characterizes conditional dependence after controlling for the remaining variables. In fact, we have that the elements Λ_{ij} are related to partial correlations through the following identity $\rho_{i,j} = -\frac{\Lambda_{i,j}}{\sqrt{\Lambda_{i,i}\Lambda_{j,j}}}$, so that zeros in the off-diagonal entries of Λ correspond to conditional indepen-

dence restrictions (Pourahmadi 2013). We therefore use the concentration matrix, together with the associated partial correlations, to define an undirected conditional independency graph.

We first introduce our proposed 2PMMM which generalize the tradition mixed membership model using joint distribution specified by Bayesian network. Let $\Pi \in \mathbb{R}^{N \times K}$ collect the unit-specific mixed-membership vectors $\pi_i \in \mathbb{R}^K$, where N is the number of units and K is the number of latent groups. The matrix Π is the main object of interest in this paper. Conditional on Π , the adjacency $A \in \mathbb{R}^{N \times N}$ is generated from a Bernoulli model with edge-probability matrix $-\Theta \Pi B \Pi' \Theta$, where $B \in \mathbb{R}^{K \times K}$ is the group-level connectivity matrix and $\Theta \in \mathbb{R}^{N \times N}$ captures node-specific heterogeneity. The diagonal entries of B govern within-group link probabilities, whereas the off-diagonal entries govern between-group link probabilities; in the assortative case, within-group probabilities are larger than between-group probabilities. This path corresponds to a standard mixed-membership network model (Fan et al. 2022, Jin et al. 2024). Figure 1a illustrates that the resulting adjacency matrix exhibits overlapping structure when units have mixed memberships. Under pure membership, the model reduces to a block structure without overlapping components.

The second path introduces node-specific sociability into the intensity of conditional dependence. We have sociability structure $U \in \mathbb{R}^{N \times N}$ which is a diagonal matrix. Each element u_i stands for sociability for i . For example, individuals from more socially connected families may have higher sociability, while individuals from



(a) The first path governs the sparsity pattern. Because units can have mixed memberships, the resulting network exhibits overlap between latent groups. (b) The second path governs connection intensity. Higher latent sociability leads to stronger interactions, while lower latent sociability leads to weaker interactions. (c) The adjacency W , generated from $\mathbb{P}(W, A)$, induces a network with the sparsity structure shown in panel (a) and the connection-weight structure shown in panel (b).

Figure 1

less socially connected families may have lower sociability. Similarly, regions with favorable harbor conditions may depend on external trade than inland regions. This parameter measures the latent tendency of unit i to form strong conditional dependence with other units. As illustrated in Figure 1b, the connection intensity between two units is determined by the interaction of their sociability parameters: pairs of highly sociable units generate larger partial-correlation weights, pairs of low-sociability units generate smaller weights, and mixed pairs generate intermediate weights.

The two paths jointly determine weighted network W . The first path governs the sparsity pattern, while the second path governs the intensity of nonzero interactions. Conditional on the adjacency matrix A , the weight of each existing link between units i and j is drawn from a beta distribution whose parameters are $u_i u_j \in \mathbb{R}$ and

$\sigma_{ij} \in \mathbb{R}$. The product $u_i u_j$ measures pairwise latent interaction intensity, and σ_{ij} acts as a pair-specific normalization parameter. Larger values of $u_i u_j$ generate stronger conditional dependence, reflected in larger entries of W , while the beta specification restricts the weights to $[0, 1]$. We point out that, in our framework, we impose the weight to be positive, which implies that the partial correlations of the model are non-negative. It is possible to relax it by using a simple transformation in Beta. However, we keep the non-negative assumption here for brevity because [Barigozzi & Brownlees \(2019\)](#) and [Brownlees et al. \(2018\)](#) documented that the fraction of negative partial correlations estimated in network models for economic panels is negligible. Figure 1c illustrates the concentration matrix generated by the 2PPCN model.

Following the principles of Bayesian belief networks ([Koller & Friedman 2009](#)), the 2PPCN can be represented by a sequence of simplified conditional distribution as follows:

$$\begin{aligned} & \mathbb{P}(W, A | B, \Theta, \Pi, \mathbf{U}) \\ &= \left\{ \prod_{i,j}^N \mathbb{P}(W_{ij} | A_{ij}, \sigma_{ij}, u_i, u_j) \right\} \left\{ \prod_{i,j}^N \mathbb{P}(A_{i,j} | B, \pi_i, \pi_j, \Theta_i, \Theta_j) \right\}. \end{aligned} \quad (1)$$

Each conditional distribution in Equation (1) is specified by a specific conditional distribution in two paths above which is formally defined in Definition 1.

Remark 1 (Conditional independence assumptions). *The factorization in Equation (1)*

follows from Bayes' rule together with the conditional independence assumptions encoded by the proposed probabilistic graphical model. Specifically, we assume that, conditional on its direct parent variables, each component is independent of the remaining variables in the model. These assumptions can be written as

$$\begin{aligned}\mathbb{P}(W \mid A, \Pi, B, \Theta, \mathbf{U}, \sigma) &= \mathbb{P}(W \mid A, \sigma, \mathbf{U}) = \prod_{i,j=1}^N \mathbb{P}(W_{ij} \mid u_i, u_j, \sigma_{ij}, A_{ij}), \\ \mathbb{P}(A \mid \Pi, B, \Theta, \mathbf{U}, \sigma) &= \mathbb{P}(A \mid \Pi, B, \Theta) = \prod_{i,j=1}^N \mathbb{P}(A_{ij} \mid \pi_i, \pi_j, B, \Theta_i, \Theta_j).\end{aligned}$$

These restrictions imply that W depends on the remaining model components only through $(A, \sigma, \mathbf{U})$, the network A are conditionally independent of the other parameters given (Π, B, Θ) . Under these assumptions, the joint distribution admits the factorization stated in Equation (1).

Eventually, the standard 2PPCN is formally defined as follows:

Definition 1. (*Two-Path Mixed Membership Model*). Given normalized parameter σ , unit-specific heterogeneity matrix Θ and the community-specific probability matrix B , unit-specific sociability matrix $\mathbf{U}$, the concentration matrix is generated sequentially in

the following way:

$$A_{ij} \mid \pi_i, \pi_j, B, \Theta_i, \Theta_j \sim \text{Bernoulli}(\Theta_i \pi_i B \pi_j' \Theta_j) \quad \text{for } i, j = 1, \dots, N.$$

$$W_{ij} \mid A_{ij}, \sigma_{ij}, u_i, u_j = \begin{cases} \text{Beta}(u_i u_j, \sigma_{ij}) & \text{if } A_{ij} = 1 \\ 0 & \text{if } A_{ij} = 0 \end{cases} \quad \text{for } i, j = 1, \dots, N.$$

The following theorem characterizes the expectation of the concentration matrix Λ under the 2PPCN model specified in Definition 1. This expectation corresponds to the population object in Equation (1).

It is convenient to write

$$\mathcal{G} = W \sim \mathbb{P}(W, A \mid B, \Theta, \Pi, \mathbf{U})$$

Theorem 1. *Assuming a 2PPCN as defined in Definition 1 under the assumption that Π is a set of fixed parameters. Then, we have that:*

$$\Omega_W = \mathbb{E}(W) = \mathbf{U} \Theta \Pi B \Pi' \Theta \mathbf{U}.$$

Theorem 1 characterizes the population concentration matrix W generated by the proposed 2PPMM model. Because W is a linear function of W , $\mathbb{E}(\Lambda)$ can be derived

from $\mathbb{E}(W)$ at once. This characterization provides the theoretical foundation for the mixed-membership estimation algorithm developed below.

Finally, we introduce a model in which the partial correlation structure of the random vector Y_t is determined by a latent 2PMMN $\mathcal{G}$. We employ the function used in [Brownlees et al. \(2022\)](#) such that the concentration Λ is a function of the Laplacian L of the latent graph W such that

$$\Lambda = \frac{1}{\underline{\sigma}^2} I_N + \frac{\phi}{\underline{\sigma}^2} L. \quad (2)$$

where $L = D_W^{-1/2} W D_W^{-1/2}$ and W is defined in Definition 1, $D_W = \text{diag}\left(\sum_{j \neq i}^N W_{ij}\right)_{i=1}^N$, $\underline{\sigma}^2$ is the network variance and ϕ is a dependence parameter. Since Λ is a function of W , they share the same distribution. It is guaranteed that the concentration Λ is symmetric and positive definite.

Definition 2. (*Two-Path Mixed Membership Partial Correlation Network Model*). Let $\mathcal{G} \sim \mathbb{P}\left(W, A \middle| B, \Theta, \Pi, U\right)$ be a 2PPMM as defined in Definition 1. Let Λ be the $N \times N$ concentration matrix corresponding to the random graph $\mathcal{G}$ as defined in Definition 2. In a two-path mixed membership partial correlation model, the N -dimensional stationary stochastic process $\{Y_t\}_{t=1}^T$ has mean zero and covariance matrix, $\Sigma = \Lambda$.

Proposition 1 gives the sufficient condition for 2PMMPCN model to be symmetric and positive definite.

Proposition 1. *Consider the 2PPCN as defined in Definition 2 and assume that $\underline{\phi} \in [0, 1]$. Then, the 2PPCN is symmetric and positive definite.*

2.2 Algorithm

In this subsection, we propose an algorithm to estimate the mixed membership structure in a panel generated through the 2PPCN model. To this end, we extend the SCORE algorithm of Jin (2015) and its mixed-membership variant, Mixed-SCORE, proposed in Jin et al. (2024).

We first estimate the number of latent groups in the panel using an information-criterion method. Given the panel data $\{Y_t\}_{t=1}^T$ and the estimated number of communities $\hat{K}$, the first step is to estimate the covariance matrix of the panel innovations. When $T > N$, we use the unregularized sample covariance estimator, $\hat{\Sigma} = \frac{\mathbf{Y}'\mathbf{Y}}{T}$. When $T < N$, $\hat{\Sigma}$ is singular and cannot be inverted directly. Extending the procedure to this high-dimensional case requires replacing the unregularized inverse covariance estimator with a regularized precision matrix estimator, such as the graphical Lasso, CLIME. Next, we calculate the sample concentration matrix $\hat{\Lambda} = \hat{\Sigma}^{-1}$. We set all diagonal entries of $\hat{\Lambda}$ to 0. Then, we normalize $\hat{\Lambda}$ to obtain its symmetric normalized Laplacian: $\hat{L} = \hat{D}^{-1/2}\hat{\Lambda}\hat{D}^{-1/2}$, where $\hat{D}$ is the diagonal matrix of row (or column) sums of $\hat{\Lambda}$. This step follows the standard normalization in spectral graph theory (von Luxburg 2007). The next step is the spectral decomposition of graph Laplacian. Let $\hat{\mathbf{X}} \in \mathbb{R}^{N \times K}$ contain the eigenvectors of $\hat{L}$ corresponding to its

K largest eigenvalues. The rows of $\hat{\mathbf{X}}$ can be interpreted as a rotation of $\frac{1}{2}\hat{\mathbf{U}}\hat{\mathbf{\Theta}}\hat{\mathbf{\Pi}}$ with an unknown rotation matrix $\hat{P} \in \mathbb{R}^{K \times K}$. Next, for each node $i = 1, \dots, N$, we calculate $\hat{r}_i = \left(\frac{\hat{\mathbf{x}}_{i,2}}{\hat{\mathbf{x}}_{i,1}}, \dots, \frac{\hat{\mathbf{x}}_{i,K}}{\hat{\mathbf{x}}_{i,1}} \right) \in \mathbb{R}^{K-1}$, where $\hat{\mathbf{x}}_{i,k}$ is the k -th component of the i -th row of $\hat{\mathbf{X}}$. This step projects the mixed-membership structure onto a $(K-1)$ -dimensional simplex, eliminating the influence of $\hat{\mathbf{U}}$ and $\hat{\mathbf{\Theta}}$ while preserving relative positions of all $\hat{\pi}_i$'s in a K -dimensional simplex. Next, we identify the simplex corner on which $\hat{r}_i$'s locate. We locate K anchor points $\{\hat{c}_k\}_{k=1}^K$, where $\hat{c}_k \in \mathbb{R}^K$ serve as approximate basis vertices spanning the simplex containing $\hat{r}_i$'s. We next compute weights $\hat{w}_i \in \mathbb{R}^K$ for $i = 1, \dots, N$ such that $\hat{r}_i = \sum_{k=1}^K \hat{w}_{i,k} \hat{c}_k$, where the weights form a convex combination. Finally, we recover the mixed memberships by normalizing each $\hat{w}_i$ appropriately (using the normalization scheme in mixed-SCORE) to obtain the estimated mixed membership vector $\hat{\pi}_i$.

Our proposed PartialCorr-mixed-SCORE differs from SCORE and mixed-SCORE in the following ways. First, the input data type is different. SCORE and mixed-SCORE operate on binary adjacency matrices, where entries are either 0 or 1. In contrast, PartialCorr-mixed-SCORE takes panel data as input, which can contain continuous observations over time. Second, SCORE is designed for networks whose adjacency matrix A follows a Bernoulli distribution with parameters $\Theta Z B Z' \Theta$, i.e., $A \sim \text{Bernoulli}(\Theta Z B Z' \Theta)$ where Z contains pure community labels. Mixed-SCORE generalizes this to a mixed membership setting: $A \sim \text{Bernoulli}(\Theta \Pi B \Pi' \Theta)$ where Π contains the mixed membership structure. In contrast, PartialCorr-mixed-SCORE

Algorithm 1 PartialCorr-mixed-SCORE

Input: Panel data $\{Y_t\}_{t=1}^T$ and estimated number of communities $\hat{K}$.

- 1: Compute the sample covariance matrix $\hat{\Sigma} = \frac{\mathbf{Y}'\mathbf{Y}}{T}$ and the sample concentration matrix

$$\hat{\Lambda} = \hat{\Sigma}^{-1}.$$

- 2: Set the diagonal entries of $\hat{\Lambda}$ to zero.
3: Compute the normalized Laplacian

$$\hat{L} = \hat{D}^{-1/2} \hat{\Lambda} \hat{D}^{-1/2},$$

where $\hat{D}$ is the diagonal degree matrix associated with $\hat{\Lambda}$.

- 4: Let $\hat{X} = [\hat{\xi}_1, \dots, \hat{\xi}_K] \in \mathbb{R}^{N \times K}$ be the matrix of the first K eigenvectors of $\hat{L}$ corresponding to its K largest eigenvalues.
5: Set $S = \log(N)$ and construct the entry-wise ratio matrix

$$\hat{R}_{ik} = \text{sign} \left(\frac{\hat{X}_{i,k+1}}{\hat{X}_{i,1}} \right) \min \left\{ \left| \frac{\hat{X}_{i,k+1}}{\hat{X}_{i,1}} \right|, S \right\},$$

for $1 \leq i \leq N$ and $1 \leq k \leq K-1$.

- 6: Find K corner points $\{\hat{v}_1, \dots, \hat{v}_K\}$ from the rows of $\hat{R}$ using a corner-search algorithm.
7: For each $i = 1, \dots, N$, solve

$$\hat{R}_i = \sum_{k=1}^K \hat{w}_{ik} \hat{v}_k, \quad \sum_{k=1}^K \hat{w}_{ik} = 1.$$

- 8: For $1 \leq i \leq N$ and $1 \leq k \leq K$, compute

$$\hat{\pi}_{ik}^* = \max \left\{ 0, \frac{\hat{w}_{ik}}{\hat{b}_{1k}} \right\},$$

where

$$\hat{b}_{1k} = \left[\hat{\lambda}_1 + \hat{v}_k' \text{diag}(\hat{\lambda}_2, \dots, \hat{\lambda}_K) \hat{v}_k \right]^{-1/2}.$$

- 9: Normalize

$$\hat{\pi}_i = \frac{\hat{\pi}_i^*}{\|\hat{\pi}_i^*\|_1}, \quad i = 1, \dots, N.$$

Output: Estimated mixed membership matrix $\hat{\Pi} \in \mathbb{R}^{N \times K}$.

applies to panel data generated from panel generating process such that $Y_t = \epsilon_t$ where $\epsilon_t \sim \mathcal{N}(0, \Lambda^{-1})$ and $\Lambda \sim 2PPCN$.

2.3 Theory

This section establishes theoretical guarantees for the proposed PartialCorr-Mixed-SCORE algorithm when applied to large panels of time series. The main results provide bounds on the mean squared error between the estimated mixed-membership vector $\hat{\pi}_i$ and its population counterpart π_i . Because the model allows mixed membership, the usual misclassification rate used in pure-membership clustering is not directly applicable. Instead, we evaluate estimation accuracy through the MSE between the estimated and population membership vectors. The theoretical analysis proceeds in two steps. First, we show that PartialCorr-Mixed-SCORE recovers the mixed-membership structure at the population level, where the population graph is defined as in Theorem 1. Second, we derive error bounds for the MSE between the membership vectors estimated from the sample concentration matrix and their population counterparts. The sample concentration matrix is obtained from the panel data. The identification of the model is discussed in Section B.1 of the supplementary material.

The following Lemma establishes that there exists a rotation matrix P such that $\mathbf{X} = \left(\mathcal{D}_\Omega^{-1/2} \mathbf{U} \Theta \Pi \right) P$, where $\mathbf{X}$ denotes the matrix of eigenvectors corresponding to the top K eigenvalues of $\mathcal{L}$ and $\mathcal{D}_\Omega = \text{diag} \left(\sum_{j \neq i}^N \Omega_{W,ij} \right)_{i=1}^N$. Since $\mathbf{\Lambda} = \frac{1}{\underline{\sigma}^2} I_N + \frac{\phi}{\underline{\sigma}^2} \mathcal{L}$,

then the eigenvector space of Λ is the same as that of $\mathcal{L}$ corresponding to respective K largest eigenvalues. We study eigenvector space of Λ by studying that of $\mathcal{L}$. This result is crucial for estimating the latent mixed membership matrix Π in the 2PPCN model. Since Π is unobservable and the observed data consist only of the panel Y_t , we must recover information about Π indirectly. The key insight provided by Lemma 1 is that $\mathbf{X}$, which can be computed from observable quantities (i.e., the sample partial correlation network), encodes the structure of Π up to an orthogonal transformation. This relationship enables us to estimate Π by analyzing the geometry of $\mathbf{X}$ and applying techniques such as those developed in the SCORE and mixed-SCORE frameworks (Jin 2015, Jin et al. 2024).

Lemma 1. *Let the 2PPCN defined in Definition 1 satisfying Assumption 1 and let $\lambda_1, \dots, \lambda_K$ and $\mathbf{x}_1, \dots, \mathbf{x}_K$ denote the top K eigenvalues and corresponding eigenvectors of $\mathcal{L} = (\mathcal{D}_\Omega^{-1/2} \mathbf{U} \Theta \Pi B \Pi' \Theta \mathbf{U} \mathcal{D}_\Omega^{-1/2})$, ordered from largest to smallest. Then, there exists a $K \times K$ non-singular real matrix P such that $(\mathcal{D}_\Omega^{-1/2} \mathbf{U} \Theta \Pi) P = \mathbf{X}$, where $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_K]'$. Moreover, the matrix P is unique once $\mathbf{X}$ is fixed.*

Lemma 1 paves the way for establishing the consistency of the proposed algorithm.

2.3.1 Consistency in Population

The population analysis in this subsection heavily borrows insights from Jin et al. (2024). We begin by analyzing the consistency of the PartialCorr-Mixed-SCORE algorithm under the 2PPCN model at the population level. It corresponds to the

exact recovery of Π in $\mathbb{E}(\Lambda)$ under the proposed algorithm. We make the following assumption.

Assumption 1. *Given 2PPCN, define a $K \times K$ matrix $G = K \left\| d_{\Omega}^{-1/2} u \theta \right\|^{-2} \left(\frac{1}{4} \Pi' \Theta^2 U^2 \mathcal{D}_{\Omega}^{-1} \Pi \right)$. There exists a constant C such that $\left(\mathcal{D}_{\Omega}^{-1/2} \right)_{max} \leq C$, $\Theta_{max} \leq C$, $U_{max} \leq C$, $\|B\|_{max} \leq C$, $\|G\| \leq C$ and $\|G^{-1}\| \leq C$.*

Assumption 1 is to impose upper bounds on the hyper-parameters in the 2PPCN model, which are used for establishing Lemma 2 presented below.

Before proceeding, we introduce additional notation. Let $v_i \in \mathbb{R}^{K-1}$ denote a basis vector of the simplex $\mathcal{S}$, and assume that $\mathcal{S}$ is spanned by these vertices. Let $\mathbf{X}_i \in \mathbb{R}^K$ collect the entries of the eigenvectors associated with the largest K eigenvalues for unit i . Denote by $\mathbf{X}_{i,1}$ its first component and by $\mathbf{X}_{i,-1} \in \mathbb{R}^{K-1}$ the remaining components. Define the SCORE-ratio vector $r_i = \mathbf{X}_{i,-1} / \mathbf{X}_{i,1} \in \mathbb{R}^{K-1}$. The matrix $R = (r_1, \dots, r_N)' \in \mathbb{R}^{N \times (K-1)}$ is referred to as the SCORE-R matrix throughout the paper.

Lemma 2. *Assuming a 2PPCN in Definition 1 satisfying Assumption 1 and 2 with $\mathcal{L}$ as defined in Definition 2. If B is non-singular and $B \left(\mathcal{D}_{\Omega}^{-1} \Pi' \Theta^2 U^2 \Pi \right)$ is irreducible, then, the following statements are true:*

1. *All entries in first orthonormal eigenvector $\mathbf{X}_1$ of $\mathbf{X}$ are strictly positive.*
2. *There exists a K -vertex simplex $\mathcal{S} \in \mathbb{R}^{K-1}$ which is spanned by $v_1, \dots, v_K$.*

3. All r_i 's locate on S and pure membership r_i on one of the vertices of S for $i = 1, \dots, N$.
4. There exists a barycentric coordinate w_i of r_i in S such that $w_i = (\Pi_i \otimes b_1) / \|\Pi_i \otimes b_1\|_1$ where $b_{1,k} = \left(\lambda_1 + v'_k \text{diag}(\lambda_2, \dots, \lambda_K) v_k \right)^{-1/2}$, λ_k is k -th eigenvalue of $\mathcal{L}$ and $\otimes$ is element-wise multiplication.

Lemma 2 shows that the eigenvector representation $\mathbf{X}_i$ for each unit i can be transformed into a SCORE-ratio vector r_i that lies in a simplex. This transformation corresponds to the SCORE normalization of Jin (2015). At the same time, the mixed-membership vectors in Π lie in a simplicial cone with K supporting rays. Pure nodes lie exactly on one of these supporting rays, corresponding to their group membership, whereas mixed-membership nodes lie in the interior of the cone. The Mixed-SCORE framework of Jin et al. (2024) exploits this geometry to recover Π from the eigenvector matrix $\mathbf{X}$ of the graph. Lemma 2 is a direct extension of Lemma 2.1 in Jin et al. (2024); it shows that the geometric and spectral relationships among Π , the simplex, and $\mathbf{X}$ continue to hold under the 2PPCN model. This result is sufficient to show that Π can be recovered under the 2PPCN model by leveraging this relationship, which implies the consistency of the proposed algorithm on the population graph $\mathcal{L}$. Furthermore, we remark that $\mathcal{L} = (\mathcal{D}_\Omega^{-1/2} \mathbf{U} \Theta \Pi) B (\Pi' \Theta \mathbf{U} \mathcal{D}_\Omega^{-1/2})$ can be seen as a special case of $\Omega = (\tilde{\Theta} \Pi) B (\Pi' \tilde{\Theta})$ in Jin et al. (2024) if we combine $(\mathcal{D}_\Omega^{-1/2} \mathbf{U} \Theta)$ into $\tilde{\Theta}$. As a result, the proof of the simplex structure remains essentially unchanged.

2.3.2 Bounding the Deviation

In this subsection, we establish a bound on the difference between the estimated mixed membership and its population counterpart. Since the observable object in our setting is the panel data $\{Y_t\}$, rather than the graph itself, and our algorithm is applied directly to the panel, we must impose appropriate distributional and mixing assumptions on the data. Next, we discuss the mixing and distributional assumptions required for the theoretical analysis. We heavily borrow insights from Assumption 2 in [Brownlees et al. \(2022\)](#). Specifically, we require conditions that ensure the behavior for isotropic random vectors of the form $\Sigma_Y^{-1/2}Y_t$, where $\Sigma_Y = \mathbb{E}(Y_t Y_t' | \mathcal{G})$. Under the simplifying assumption $Y_t = \epsilon_t$, the covariance matrix of Y_t coincides with that of the innovation process; that is, $\Sigma_Y = \Sigma$. Let $\mathcal{B}_{-\infty}^r$ and $\mathcal{B}_{r+m}^\infty$ be the σ -algebras generated by the sequences $\{\Sigma_Y^{-1/2}Y_t : -\infty \leq t \leq r\}$ and $\{\Sigma_Y^{-1/2}Y_t : r+m \leq t \leq \infty\}$, respectively. The α -mixing coefficient is then defined as $\alpha(m) = \sup_r \sup_{A \in \mathcal{B}_{-\infty}^r, B \in \mathcal{B}_{r+m}^\infty} \left| \mathbb{P}(A \cap B | \mathcal{G}) - \mathbb{P}(A | \mathcal{G}) \mathbb{P}(B | \mathcal{G}) \right|$. We impose the following assumptions on the panel data process $\{Y_t\}$:

Assumption 2. *Let $\{Y_t\}$ be a zero-mean stationary process with covariance matrix Σ_Y , we assume without conditions that:*

1. *$\{\Sigma_Y^{-1/2}Y_t\}$ is strongly mixing with coefficient satisfying $\alpha(m) \leq e^{-C_1 m^{\gamma_1}}$, where m is a positive integer and γ_1 and C_1 are positive constants.*
2. *For any vector $\mathbf{x}$ satisfying $\|\mathbf{x}\| = 1$ and for any $s > 0$, $\mathbb{P}(|\mathbf{x}' \Sigma_Y^{-1/2} Y_t| > s | \mathcal{G}) \leq$*

$e^{1-(s/C_2)^{\gamma_2}}$, for all t and C_2 and γ_2 are positive constants.

3. $\gamma < 1$ and $1/\gamma = 1/\gamma_1 + 1/\gamma_2$.

Assumption 2.1 requires that the α -mixing coefficients decay sufficiently fast. Assumption 2.2 imposes generalized exponential tails on the distribution of $\Sigma_Y^{-1/2}Y_t$ (Fan et al. 2013). Finally, Assumption 2.3 specifies a bound for γ_1 and γ_2 . Together, Assumptions 2.1–2.3 enable the use of the concentration inequality from Merlevède et al. (2011) which is instrumental in proving Theorem 2 below.

Theorem 2 establishes that the spectral norm difference between the estimated and the population eigenvector matrix is bounded with high probability. It is a direct extension to Theorem 1 in Brownlees et al. (2022). This result builds upon a bound for the discrepancy between the estimated eigenvectors of the sample covariance and its population counterpart. It also serves as a key intermediate step in the proof of Theorem 3.

Theorem 2. *Assuming a 2PPCN as in Definition 1 satisfying Assumption 1 and the panel Y_t generated from 2PPCN satisfying Assumption 2. Let $\mathbf{O}$ be a $k \times k$ orthonormal rotation matrix, $\underline{\Pi} = \arg \min_i \|\Pi_i\|^2$ for $i = 1, \dots, N$ and let $\mathcal{X}$ denotes the eigenvector matrix of $\mathbb{E}(\Lambda)$ and $\hat{X}$ the eigenvector matrix of $\hat{\Lambda}$, both corresponding to the largest K eigenvalues. Then, we have that:*

$$\|\mathbf{O}\hat{X} - \mathcal{X}\| = O_p\left(\sqrt{\frac{\log(N)}{\underline{\Pi}NB_N}} + \sqrt{\frac{N}{T}}\right).$$

The result of Theorem 2 is instrumental in establishing Theorem 3 below, where we aim to bound the deviation of the sample SCORE- R matrix from its population counterpart, where R is the stack of all r_i for $i = 1, \dots, N$ as discussed above. By leveraging the bounded difference established in Theorem 2, we next bound the deviation between population and sample R matrix, The former is constructed from $\mathcal{X}$, while the latter is constructed from $\hat{X}$.

Theorem 3. *Assuming a 2PPCN in Definition 1 satisfying Assumption 1 and the panel data Y_t satisfying Assumption 2. Let R and $\hat{R}$ respectively denote SCORE- R of $\hat{X}$ and $\mathcal{X}$ as defined in Theorem 2. Then, we have that:*

$$\|\mathbf{L}\hat{R} - R\| = O_p\left(D\sqrt{K}\left(\sqrt{\frac{\log(N)}{\Pi NB_N}} + \sqrt{\frac{N}{T}}\right)\right),$$

$$\text{where } D = \frac{\|d_\Omega^{-1/2}\theta u\| - \left(\sqrt{\frac{\log(N)}{\Pi NB_N}} + \sqrt{\frac{N}{T}}\right)}{C\left(d_\Omega^{-1/2}\theta u\right)_{\min}}.$$

Theorem 3 provides a high-probability bound on the deviation between the population SCORE- R matrix and its sample counterpart. This bound is then used as an intermediate step in deriving the membership estimation error bound for $\|\hat{\Pi} - \Pi\|$. Specifically, $\|\mathbf{L}\hat{R} - R\|$ bounds the perturbation of the SCORE point cloud, but it does not by itself imply a membership error bound. To translate this perturbation bound into an error bound for mixed-membership vectors, we also require stability of the inverse map from the SCORE coordinates to the barycentric coordinates. When

the population simplex is well conditioned and the vertex-search step recovers its vertices reliably, small perturbations in the SCORE coordinates lead to small errors in the estimated mixed-membership vectors. Given $r_i = \sum_{k=1}^K w_i(k)v_k$, the membership reconstruction map

$$r_i \mapsto w_i \mapsto \frac{w_i \oslash b_1}{\|w_i \oslash b_1\|_1} = \pi_i, \text{ where } \oslash \text{ denotes element-wise division,}$$

is Lipschitz when b_1 , as defined in Lemma 2.4, is bounded away from zero and infinity. If $(\mathbf{L}\hat{v} - v)$ is properly bounded, then, the same rate transfers from $(\mathbf{L}\hat{R} - R)$ to $(\hat{\Pi} - \Pi)$. Lemma 2.2 implies that there exists vertices $v_1, \dots, v_K \in \mathbb{R}^{K-1}$ such that each population SCORE-R point satisfies

$$r_i = \sum_{k=1}^K w_{i,k}v_k, \quad \sum_{k=1}^K w_{i,k} = 1, \quad w_{i,k} \geq 0.$$

Let denote the population simplex matrix and its transformation by

$$\mathcal{V} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ v_1 & v_2 & \cdots & v_K \end{pmatrix} \in \mathbb{R}^{K \times K} \text{ and, respectively, } \mathcal{V}w_i = \begin{pmatrix} 1 \\ r_i \end{pmatrix} \in \mathbb{R}^K.$$

The following Assumption 3-5 are standard in Mixed-SCORE-type arguments.

Assumption 3. (Well-conditioned simplex). *There exists a constant $c_{\mathcal{V}} > 0$ such that $\sigma_{\min}(\mathcal{V}) \geq c_{\mathcal{V}}$, where $\sigma_{\min}(\mathcal{V})$ denotes the smallest singular value of the matrix $\mathcal{V}$.*

Assumption 3 ensures that the map from the SCORE coordinates r_i to the barycentric coordinates w_i is well conditioned. Geometrically, the assumption rules out degeneracy of the population simplex. For instance, when $K = 3$, it requires the triangle spanned by the simplex vertices to have non-negligible area, rather than being nearly collinear.

Assumption 4. (*Stable vertex search*) . Let $\hat{v}_1, \dots, \hat{v}_K$ be the estimated vertices produced by the vertex-search step, aligned by $\mathbf{L}$. Define $\Delta_V = \max_{1 \leq k \leq K} \|\mathbf{L}\hat{v}_k - v_k\|_2$, there exists a constant $C_V > 0$ such that, with probability tending to one, $\Delta_V \leq C_V \max_{1 \leq i \leq N} \|\mathbf{L}\hat{r}_i - r_i\|_2$.

Assumption 4 controls the error from the vertex-search algorithm. While Theorem 3 bounds the perturbation of the SCORE-R matrix, an additional condition is needed to ensure that the estimated simplex vertices are close to the population vertices. Assumption 4 provides this link by requiring the vertex search error, $\max_k \|\mathbf{L}\hat{v}_k - v_k\|$, to be controlled by the SCORE-coordinate error, $\max_i \|\mathbf{L}\hat{r}_i - r_i\|$. In this sense, the assumption imposes stability of the vertex-search step.

Assumption 5. (*Stable scaling vector*). There exist constants $0 < c_b < C_b < \infty$ such that $c_b \leq \min_k b_{1,k} \leq \max_k b_{1,k} \leq C_b$. Moreover, $\|\hat{b}_1 - b_1\|_\infty = O_p\left(D\sqrt{K}\left(\sqrt{\frac{\log(N)}{\Pi NB_N}} + \sqrt{\frac{N}{T}}\right)\right)$ with D as defined in Theorem 3.

Assumption 5 imposes lower and upper bounds on b_1 . These bounds rule out degeneracy in the normalization step, ensuring that the map from barycentric coordinates

w_i to mixed-membership weights π_i is Lipschitz continuous and therefore stable.

Then, we have the following Theorem:

Theorem 4. *Assuming a 2PPCN in Definition 1 satisfying Assumption 1, 2, 4 and 5 and the panel data Y_t satisfying Assumption 3. Let π_i and $\hat{\pi}_i$ respectively denote population mixed membership structure and estimated mixed membership structure. Then, we have that:*

$$\frac{1}{N} \sum_{i=1}^N \|\hat{\pi}_i - \pi_i\|^2 = O_p \left(D^2 K \left(\frac{\log(N)}{\underline{\Pi} N B_N} + \frac{N}{T} \right) \right),$$

$$\text{where } D = \frac{\|d_{\Omega}^{-1/2} \theta u\| - \left(\sqrt{\frac{\log(N)}{\underline{\Pi} N B_N}} + \sqrt{\frac{N}{T}} \right)}{C \left(d_{\Omega}^{-1/2} \theta u \right)_{\min}}.$$

Theorem 4 establishes an upper bound on the average mixed-membership estimation error, $\frac{1}{N} \sum_{i=1}^N \|\hat{\pi}_i - \pi_i\|^2$, for the proposed PartialCorr-Mixed-SCORE algorithm. The bound implies that accurate recovery requires N/T to be sufficiently small and the number of latent groups K to remain moderate.

2.4 Discussion

The consistency analysis in this paper relies on estimating the concentration matrix using the unregularized inverse of the sample covariance matrix. This approach implicitly imposes $T > N$ so that the sample covariance matrix can be inverted reliably. When $T < N$, $\hat{\Sigma}$ can still be computed but is singular, and hence the

unregularized estimator $\hat{\Sigma}^{-1}$ is not well defined. Thus, the restriction $T > N$ reflects the particular first-stage estimator used to construct the sample concentration matrix, rather than an inherent limitation of the PartialCorr-Mixed-SCORE framework. Extending the procedure to high-dimensional settings with $T < N$ is possible by replacing $\hat{\Sigma}^{-1}$ with a regularized precision matrix estimator, such as the graphical Lasso, CLIME, nodewise Lasso.

Throughout the theoretical analysis, we impose the simplifying assumption $Y_t = \epsilon_t$. This assumption implies that the covariance matrix of the observed process is identical to the covariance matrix of the innovations. In a more general dynamic specification, the innovation would be defined as $\epsilon_t = Y_t - \mathbb{E}(Y_t \mid \mathcal{F}_{t-1})$, so that the conditional mean must be estimated before constructing the sample covariance or concentration matrix. Incorporating the resulting first-stage estimation error into the theoretical analysis is nontrivial. For this reason, we restrict attention to the baseline case $Y_t = \epsilon_t$ in the present paper.

Finally, the theoretical bound highlights the role of the number of latent groups K . Theorem 4 shows that the upper bound on $\frac{1}{N} \sum_{i=1}^N \|\hat{\pi}_i - \pi_i\|^2$ is increasing in K , implying that the estimator becomes less stable as the latent group structure becomes more complex. This dependence on K represents an important limitation of the proposed algorithm. Improving the robustness of PartialCorr-Mixed-SCORE in settings with many latent groups is therefore a useful direction for future research.

3 Simulation

3.1 Set-Up

In this section, we conduct a simulation study to examine the finite-sample performance of the proposed PartialCorr-mixed-SCORE algorithm. The simulation setup is as follows: The number of latent communities is fixed at $K=3$. The community interaction probability matrix B is specified such that $b_{i,i} = b_{j,j}$ and $b_{i,j} = b_{j,i}$ for all i, j , reflecting symmetric intra- and inter-group interaction probabilities. We then consider three parameter settings for $(b_{i,i}, b_{i,j}) : (0.25, 0.01), (0.25, 0.05)$ and $(0.5, 0.01)$. The total number of series is set to $N = 120$, with the number of pure-type series (i.e., those entirely affiliated with a single community) denoted by N_0 . Then, obviously, $N - N_0$ is the number of mixed membership series (i.e., those affiliated with more than one community). We vary N_0 across five settings: $(30, 50, 80, 100, 120)$. The sociability parameters u_i follows Uniform with parameter $[0.3, 1.0]$ for all $i = 1, \dots, N$. The node-specific heterogeneity parameters Θ_i for $i = 1, \dots, N$ are linearly spaced from 0.9 to 1.0 in increments of 0.01. For pure nodes, membership vectors $\hat{\pi}_i$ are set to one of the canonical basis vectors: $[1, 0, 0]$, $[0, 1, 0]$ or $[0, 0, 1]$. For mixed nodes, we consider two configurations: $[0.5, 0.5, 0]$ and $[0, 0.5, 0.5]$. Both ϕ and $\underline{\sigma}$ are set to one. The concentration matrix Λ is sampled according to the 2PPCN model using the hyper-parameters described above.

To induce the mixed membership structure in the panel data, we generate Y_t through

diagonal VAR(1) with coefficients drawn uniformly from $[0, 0.9]$, where ϵ_t is sampled from a multivariate normal $\mathcal{N}(0, \Sigma)$, where the concentration matrix Λ is the inverse of the covariance matrix Σ generated via the 2PPCN model, as previously described. We repeat the experiment 500 times to assess the algorithm’s finite-sample behavior. The estimation accuracy of the mixed membership vectors is evaluated using the average ℓ_2 error across all nodes: $1/N \sum_{i=1}^N \|\hat{\pi}_i - \pi_i\|^2$, with the reported value being the average over the 500 replications.

3.2 Main Result

In this subsection, we report the main results of the simulated study.

The results from applying the PartialCorr-mixed-SCORE algorithm to panel data generated under the 2PPCN model are summarized in Table 1. Overall, the algorithm performs well in finite samples. Estimation accuracy improves when the within-group pair-specific probabilities $(b_{i,i}, b_{j,j})$ are large relative to the between-group probabilities $(b_{i,j}, b_{j,i})$, as shown across the rows of Table 1.

We also observe that precision improves as the number of time periods T increases. As the ratio $b_{i,i}b_{j,j}/b_{i,j}b_{j,i}$ increases, the underlying network exhibits stronger block structure, leading to more accurate estimation of mixed memberships. Moreover, the proportion of mixed-membership nodes affects estimation performance. As this proportion increases, the recovery becomes more difficult. For a fixed total number of series N , a higher number of pure-membership series leads to more distinct block

formation and thus better estimation accuracy.

Table 1: **Estimation errors of PartialCorr-Mixed-SCORE.** Each entry reports the average estimation error over 500 simulation replications, using the metric $\frac{1}{N} \sum_{i=1}^N \|\hat{\pi}_i - \pi_i\|^2$, where $\hat{\pi}_i$ denotes the estimated mixed membership vector and π_i is the true membership vector for i .

$N = 120$						
N_0	$b_{11}, b_{22}/b_{12}, b_{21}$	$T = 100$	$T = 300$	$T = 500$	$T = 800$	$T = 1000$
30	0.25/0.01	0.190	0.168	0.158	0.150	0.145
	0.50/0.01	0.188	0.150	0.122	0.104	0.100
	0.25/0.05	0.191	0.172	0.169	0.164	0.162
50	0.25/0.01	0.214	0.187	0.170	0.154	0.142
	0.50/0.01	0.211	0.151	0.102	0.068	0.056
	0.25/0.05	0.215	0.195	0.189	0.182	0.177
80	0.25/0.01	0.252	0.195	0.154	0.103	0.090
	0.50/0.01	0.247	0.115	0.045	0.026	0.022
	0.25/0.05	0.253	0.221	0.203	0.181	0.170
100	0.25/0.01	0.275	0.200	0.124	0.069	0.056
	0.50/0.01	0.271	0.082	0.031	0.019	0.016
	0.25/0.05	0.278	0.235	0.202	0.165	0.147
120	0.25/0.01	0.300	0.186	0.089	0.048	0.040
	0.50/0.01	0.294	0.057	0.024	0.016	0.013
	0.25/0.05	0.302	0.242	0.188	0.128	0.108

These findings consistently reinforce the conclusion that estimation quality improves with clearer block structure and greater data availability. For conciseness, all graphical illustrations are reported in Section B.2 of the supplementary material.

4 Application

4.1 Learning the Mixed Memberships of U.S. States

We apply the proposed methodology to U.S. state-level payroll employment growth rates. The data consist of seasonally adjusted, annualized quarter-to-quarter growth rates in payroll employment from 1956 to 2007. Following [Hamilton & Owyang \(2012\)](#) and [Brownlees et al. \(2022\)](#), we exclude Alaska and Hawaii, yielding a panel with $N = 48$ states and $T = 207$ quarterly observations. The data are obtained from the U.S. Bureau of Labor Statistics. Our objective is to estimate the mixed-membership structure across states and compare the resulting latent affiliations with the group structures reported in earlier studies.

[Hamilton & Owyang \(2012\)](#) study business-cycle synchronicity across U.S. states using this dataset, while [Brownlees et al. \(2022\)](#) use it to detect pure-membership groups in panel data. We instead use the same panel to estimate latent mixed-membership structure. To contextualize our empirical findings, we briefly summarize the methodologies and main results of these two related studies.

[Hamilton & Owyang \(2012\)](#) apply a cross-validation procedure to estimate the number of groups and find evidence of three distinct communities in the same dataset. Following their result, we treat the number of groups as given and do not implement a separate procedure for estimating it. Methodologically, [Hamilton & Owyang \(2012\)](#) propose a Markov-switching model in which U.S. states are grouped according to

the timing of their business cycles. Using Bayesian inference, including Markov chain Monte Carlo (MCMC), they estimate the posterior distribution over group assignments. By contrast, [Brownlees et al. \(2022\)](#) assume that the partial-correlation matrix follows a stochastic block model structure and apply spectral clustering to recover discrete group memberships. Unlike the posterior-based approach of [Hamilton & Owyang \(2012\)](#), the method of [Brownlees et al. \(2022\)](#) produces point estimates of the community structure.

Table 2 reports some results from applying PartialCorr-Mixed-SCORE to the sample. [Hamilton & Owyang \(2012\)](#) identify three groups of states based on business-cycle synchronicity: oil-producing, manufacturing, and financial states. Our results are broadly consistent with this classification but allow states to have partial affiliations with multiple groups. Based on the estimated membership weights, we label the three latent groups ex post as: (1) financial and other states, (2) oil-producing and agricultural states, and (3) manufacturing states. The labels are assigned using states with pure or nearly pure memberships as benchmarks, and mixed-membership states are interpreted according to their relative weights across these benchmark groups.

Several patterns emerge from the estimated memberships. First, most states exhibit mixed memberships, suggesting that state-level employment dynamics are associated with multiple latent economic groups rather than a single group. This pattern is consistent with the diversified structure of state economies. Second, East

and West Coast states tend to have higher membership weights in the group we label as financial and service oriented, consistent with the importance of finance, technology, and professional services in these regions. Third, several Midwestern and Southern states, including Texas and Louisiana, show strong affiliations with the oil-producing and agricultural group. Maine also displays a non-negligible loading on this group, which may reflect the role of resource-based industries in its employment dynamics. Finally, traditional manufacturing centers such as Indiana and Michigan have substantial membership in the manufacturing group, while also displaying meaningful affiliation with the financial and service group. This mixed pattern is consistent with gradual structural change in these regions from industrial production toward service-oriented economic activity. For conciseness, the full estimation results are reported in Section B.3 of the supplementary material.

Table 2: Mixed Membership Estimates for U.S. States

States	Financial and Others	Oil-Producing and Agricultural	Manufacturing
Massachusetts	0.736	0	0.264
New Jersey	0.777	0	0.223
New York	0.524	0	0.476
California	0.848	0.032	0.120
Texas	0.062	0.600	0.339
Florida	0.112	0.732	0.155
Missouri	0.129	0.871	0
Maine	0.046	0.954	0
Louisiana	0.247	0.480	0.273
Michigan	0.602	0	0.398
Indiana	0.340	0	0.660

5 Conclusion

We introduce a class of partial-correlation network models in which the concentration matrix is modeled as a random matrix generated from a joint distribution specified by a probabilistic graphical model. The framework can be embedded into general panel data-generating processes through the innovation structure. We develop PartialCorr-Mixed-SCORE, a spectral algorithm for estimating the latent mixed-membership structure from panels generated by this class of models. Consistency is established when N and T are large, N/T is sufficiently small, and the number of latent groups K remains moderate. An empirical application studies mixed affiliations of U.S. states with latent industry-related groups.

Several extensions are left for future research. One direction is to adapt Mixed-SCORE so that its performance is more robust to the choice of the number of latent groups K . Another direction is to embed the innovation process ϵ_t into richer panel data-generating processes, including factor models, fixed-effects models, and autoregressive specifications. Finally, the directed acyclic graph considered in this paper is designed for the proposed 2PPCN framework. Alternative DAG could be constructed for other modeling environments, with inference and theoretical analysis carried out according to the joint distribution induced by the corresponding graph.

6 Disclosure statement

The authors declare that no conflicts of interest exists.

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